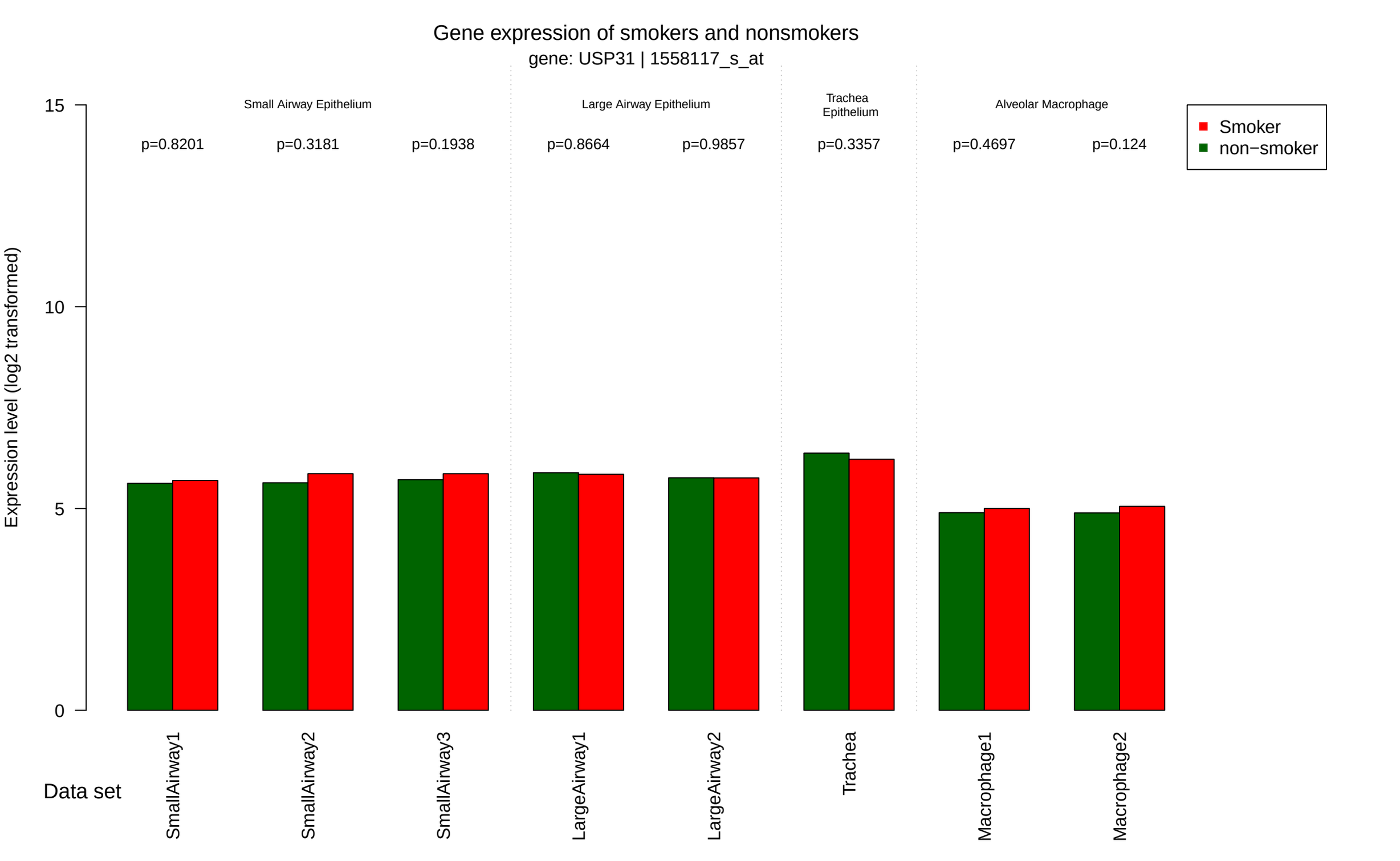
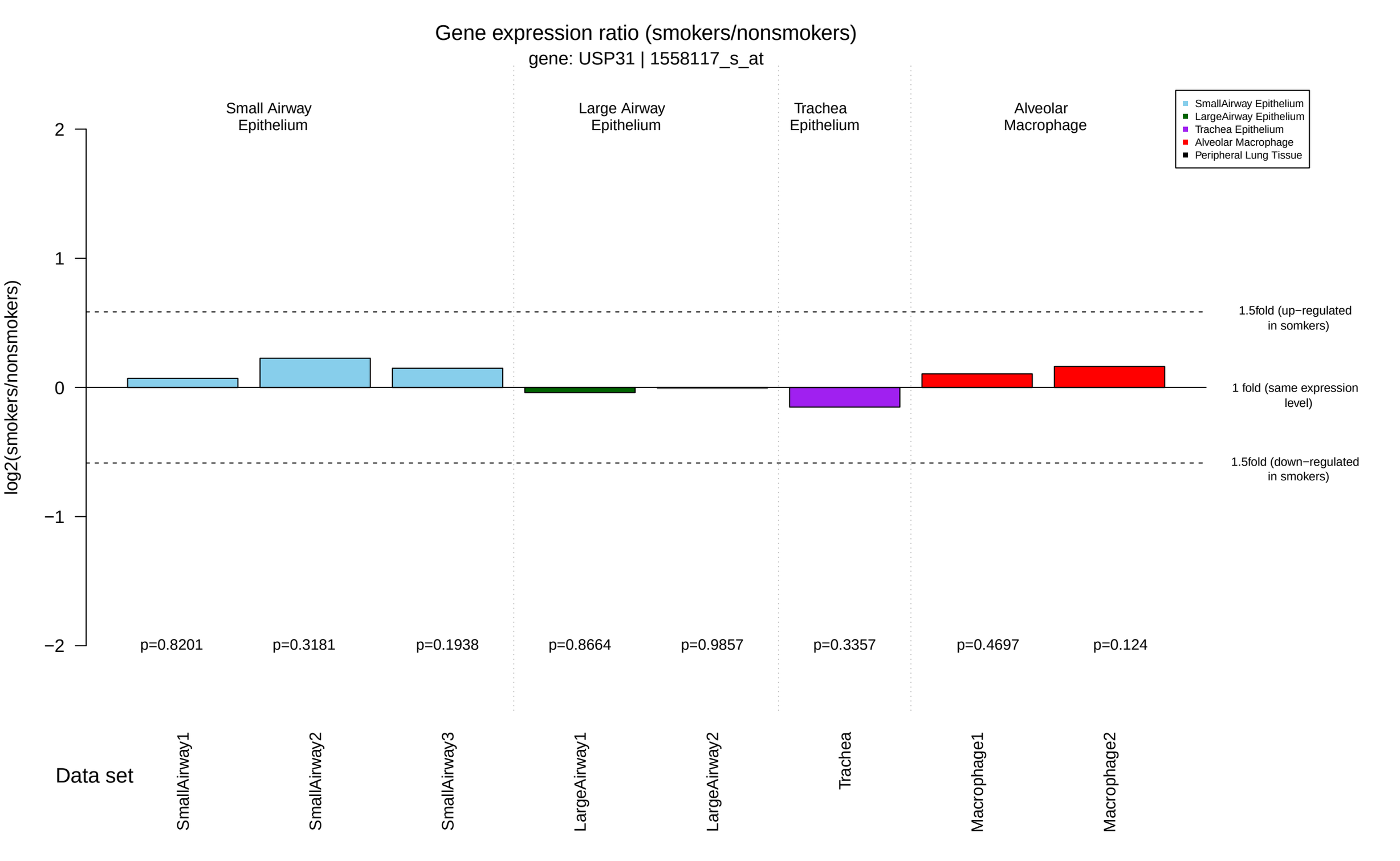
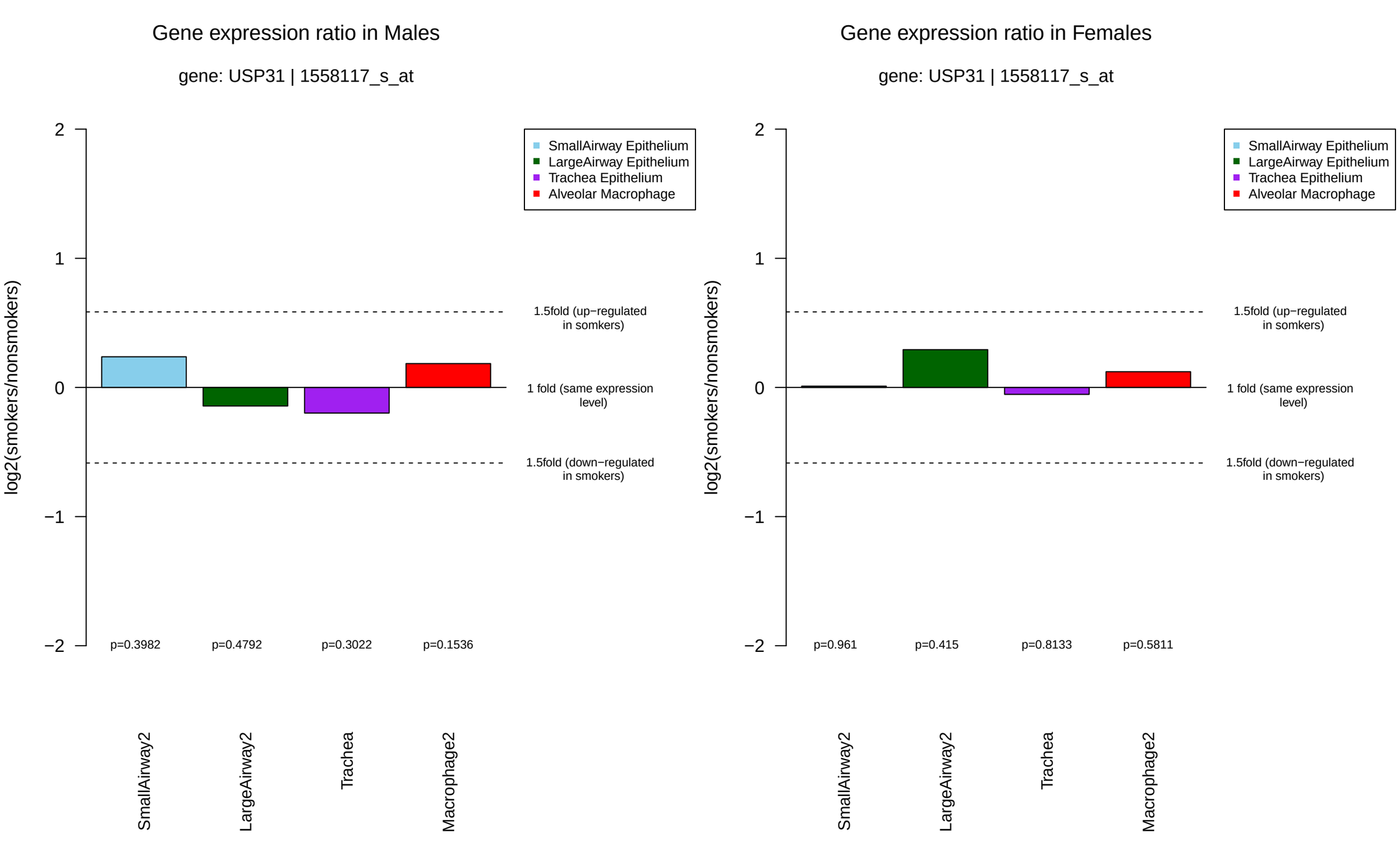
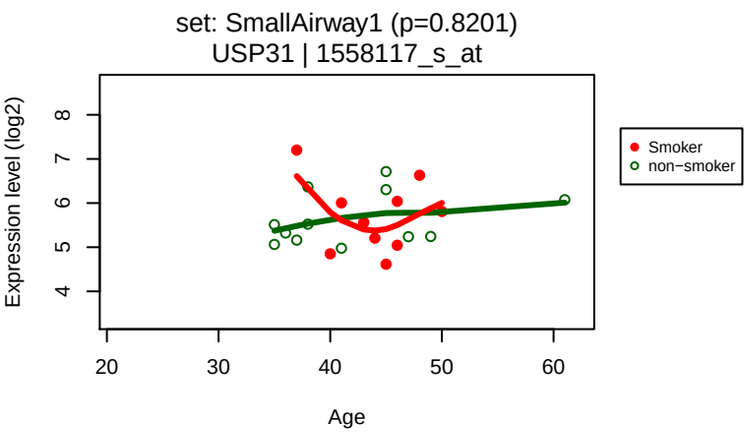
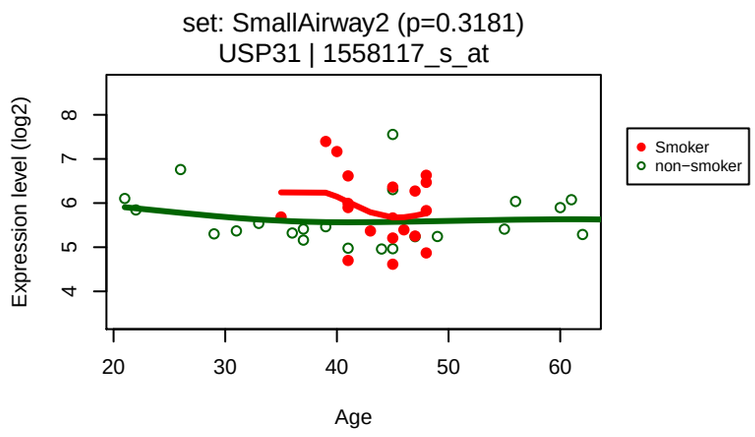
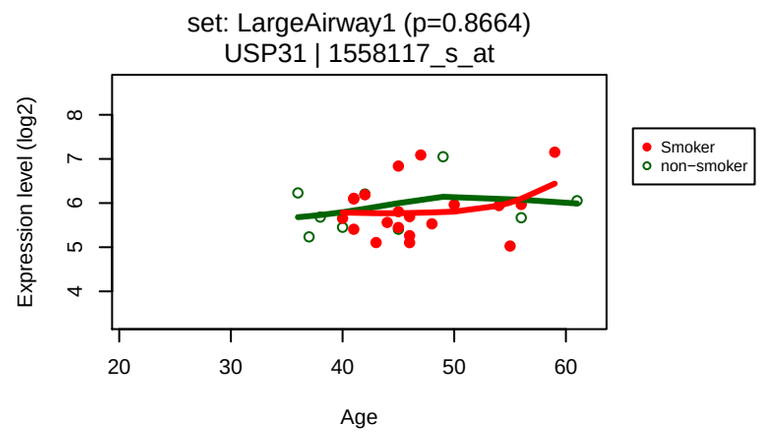

Gene expression of smokers and nonsmokers
gene: USP31 | 1558117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8201
p=0.3181
p=0.1938
p=0.8664
p=0.9857
p=0.3357
p=0.4697
p=0.124
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

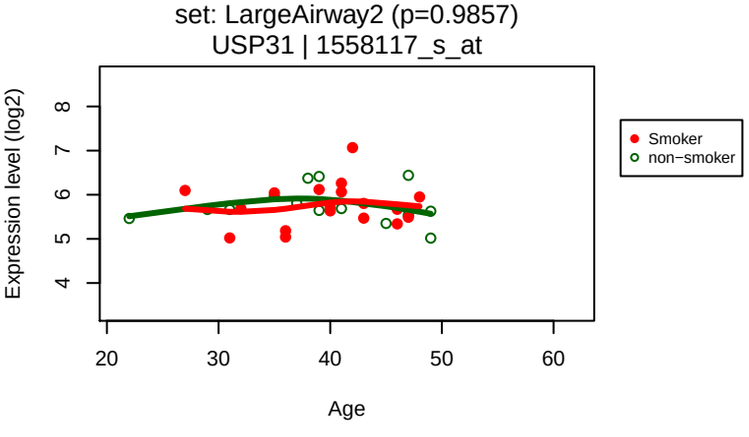

Gene expression ratio (smokers/nonsmokers)
gene: USP31 | 1558117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8201
p=0.3181
p=0.1938
p=0.8664
p=0.9857
p=0.3357
p=0.4697
p=0.124
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

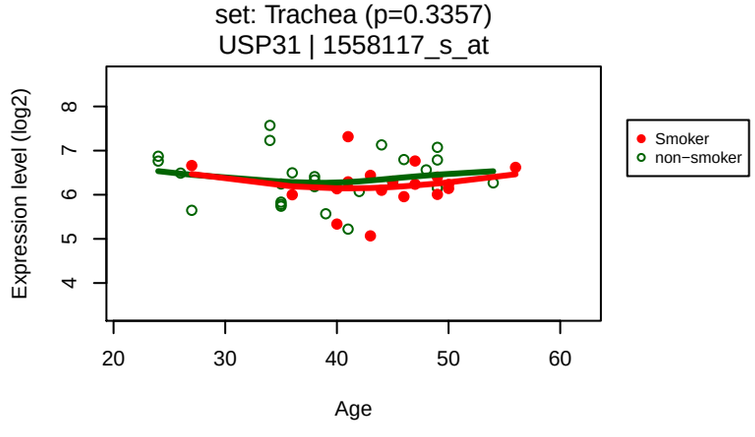

Gene expression ratio in Males
gene: USP31 | 1558117_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3982
p=0.4792
p=0.3022
p=0.1536
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP31 | 1558117_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.961
p=0.415
p=0.8133
p=0.5811
SmallAirway2
LargeAirway2
Trachea
Macrophage2

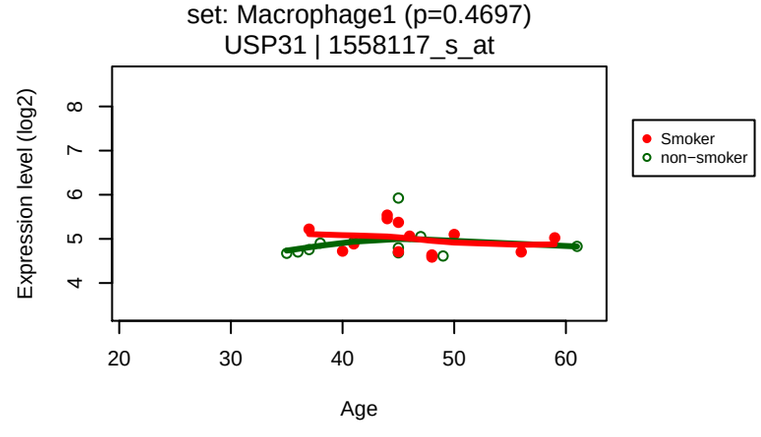

set: SmallAirway1 (p=0.8201)
USP31 | 1558117_s_at
Expression level (log2)
Age
Smoker
non−smoker

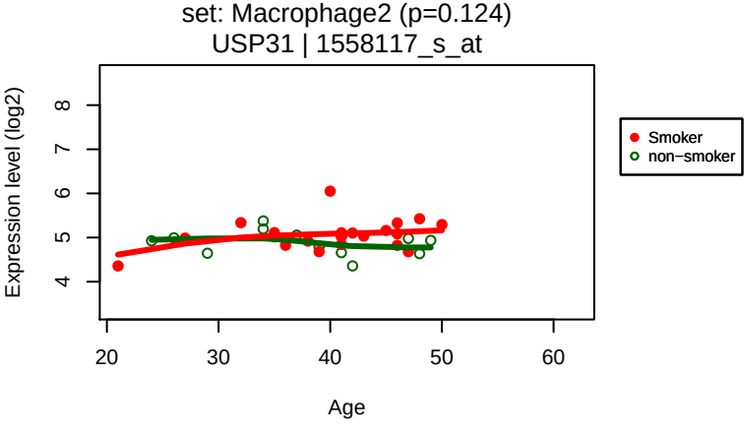

set: SmallAirway2 (p=0.3181)
USP31 | 1558117_s_at
Expression level (log2)
Age
Smoker
non−smoker

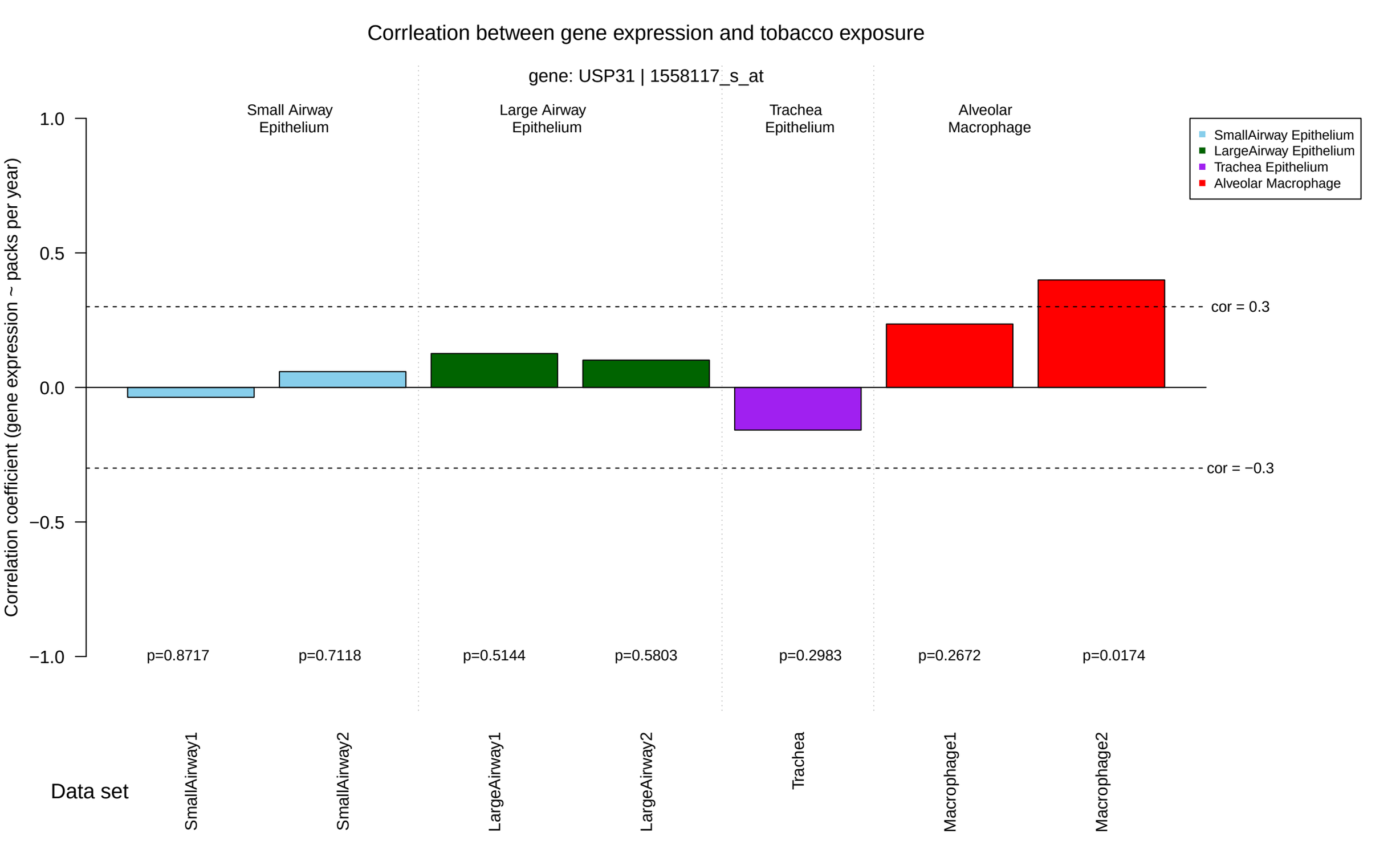

set: LargeAirway1 (p=0.8664)
USP31 | 1558117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9857)
USP31 | 1558117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3357)
USP31 | 1558117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4697)
USP31 | 1558117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.124)
USP31 | 1558117_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP31 | 1558117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8717
p=0.7118
p=0.5144
p=0.5803
p=0.2983
p=0.2672
p=0.0174
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2